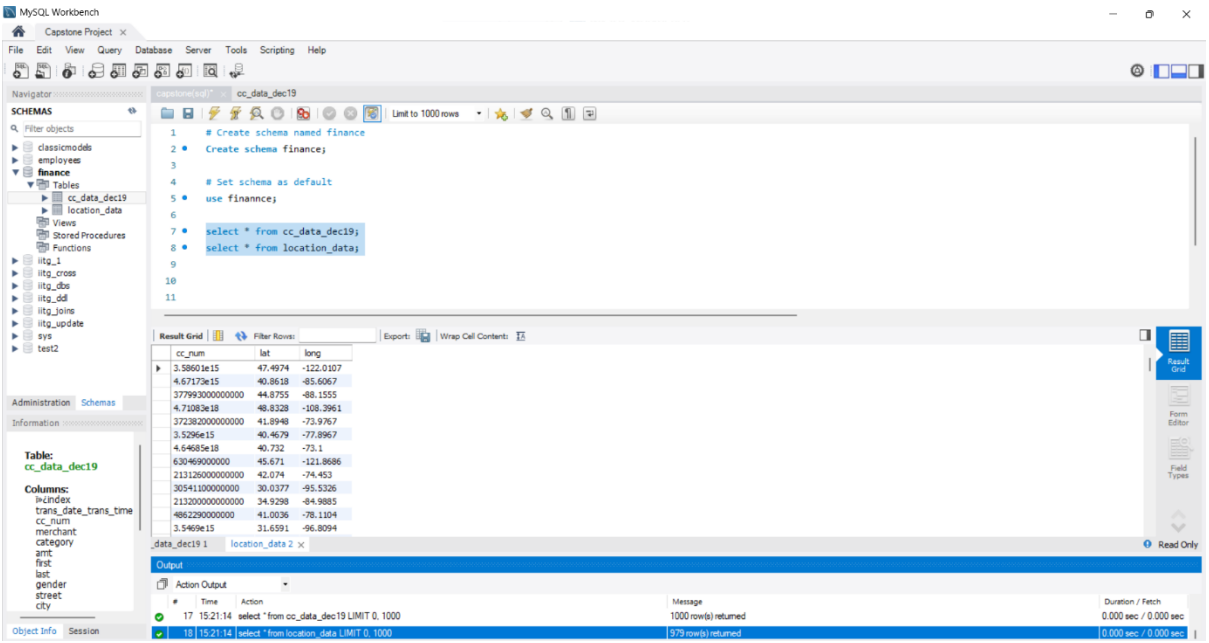
Capstone 1: SQL

Financial Fraud Detection: Anomaly Detection in Credit Card Transactions

SQL Tasks:

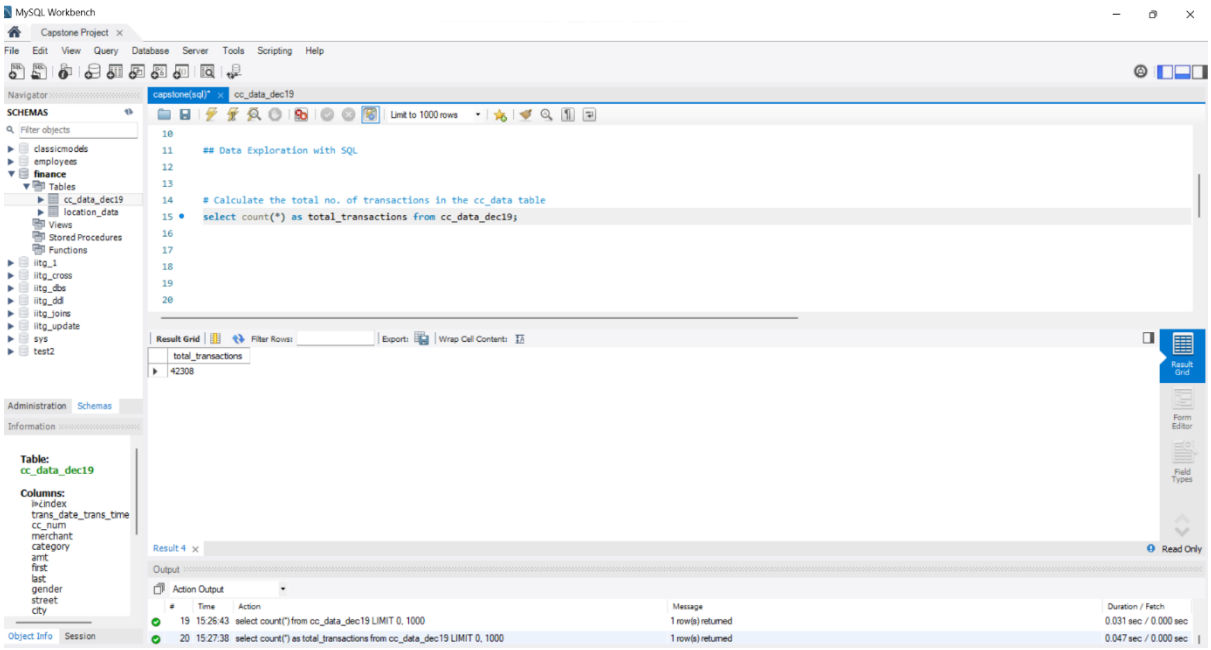
1. Data Loading:

    1.1. Create a schema named **finance**, set it as the default schema, and create tables using **cc_data_dec19.csv** and **location_data.csv**

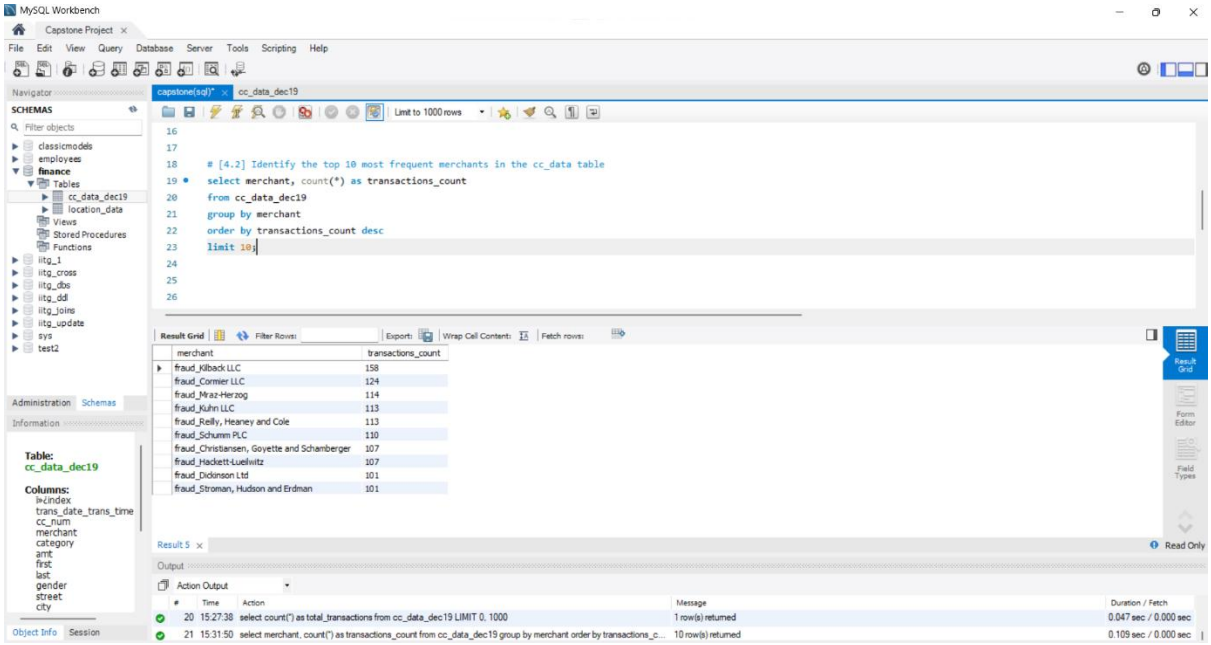


2. Data Exploration with SQL:

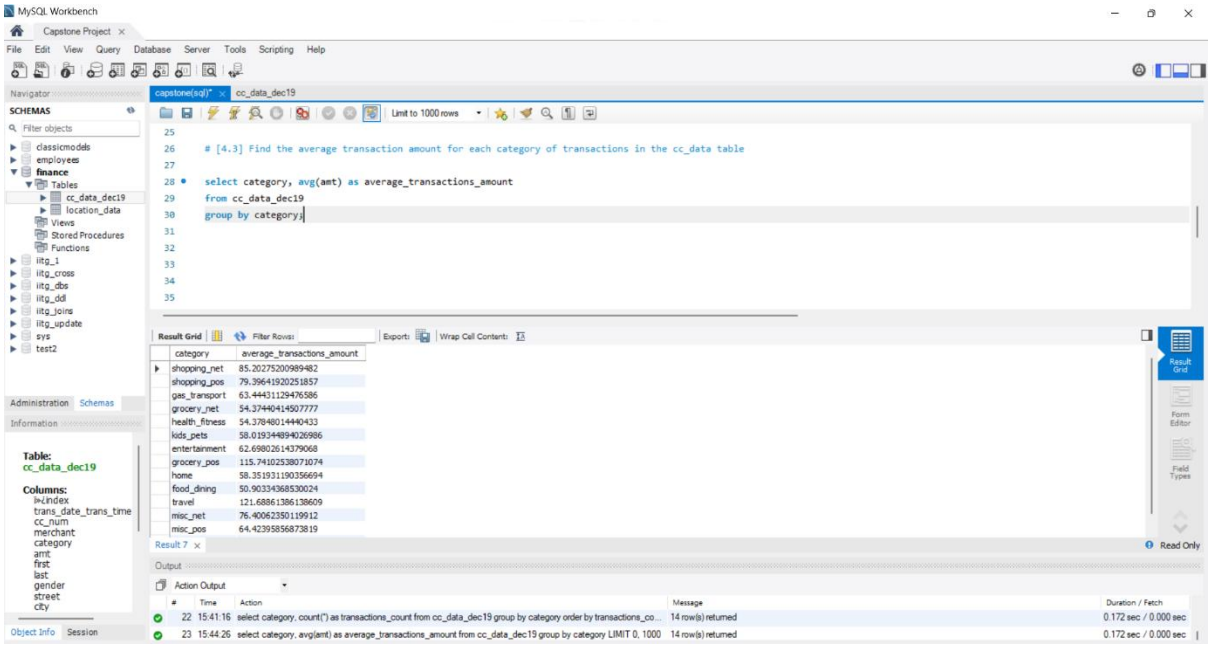
2.1. Calculate the total number of transactions in the **cc_data_dec19** table



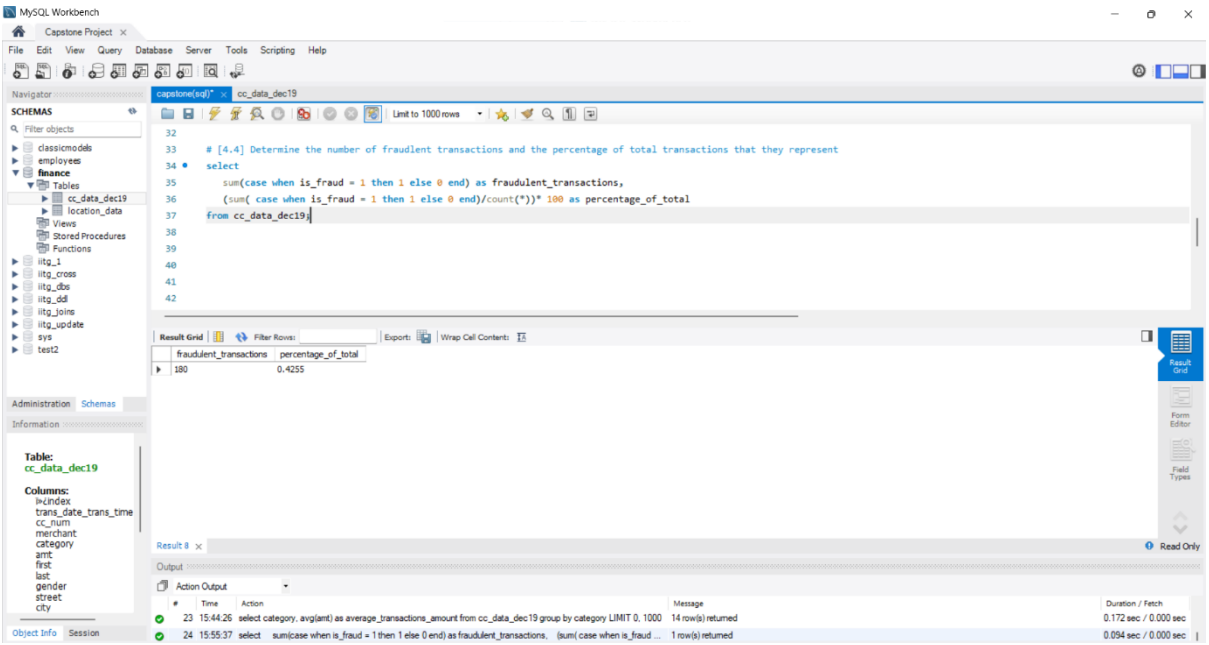
2.2. Identify the top 10 most frequent merchants in the **cc_data_dec19** table



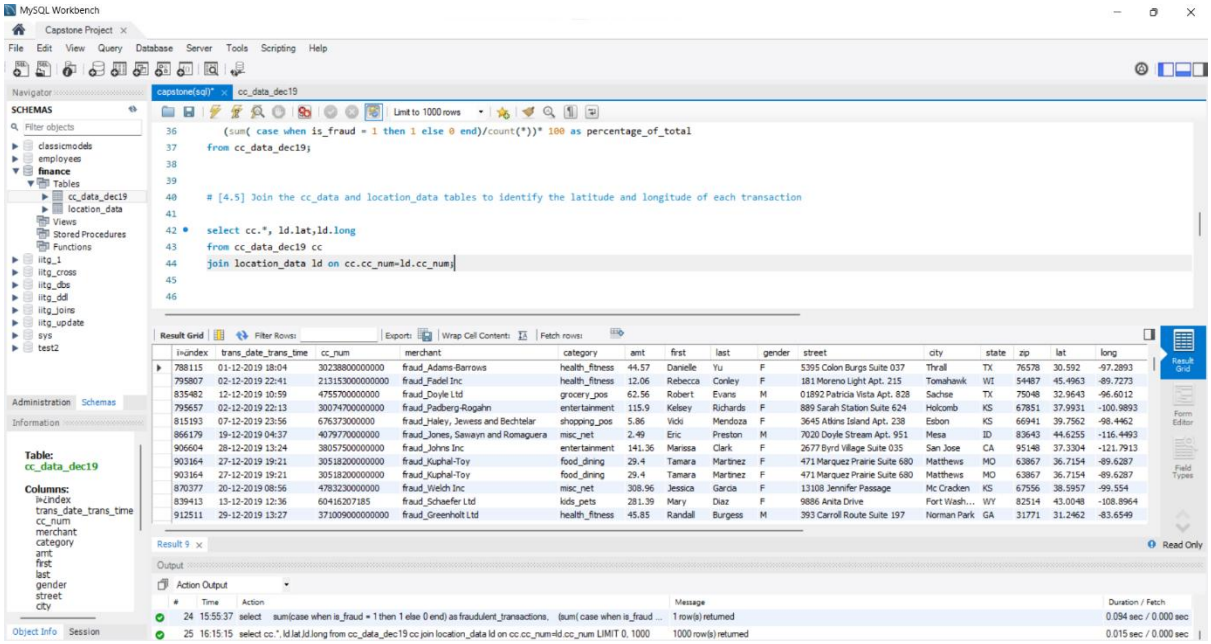
2.3. Find the average transaction amount for each category of transactions in the **cc_data_dec19** table



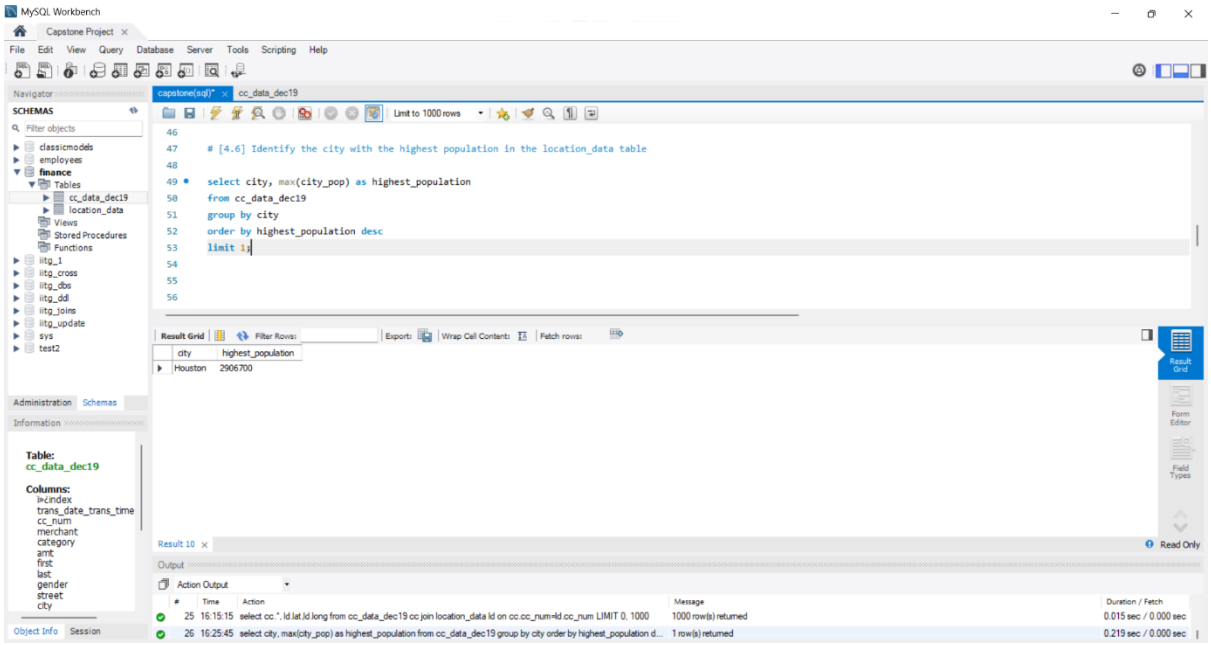
2.4. Determine the number of fraudulent transactions and the percentage of total transactions that they represent



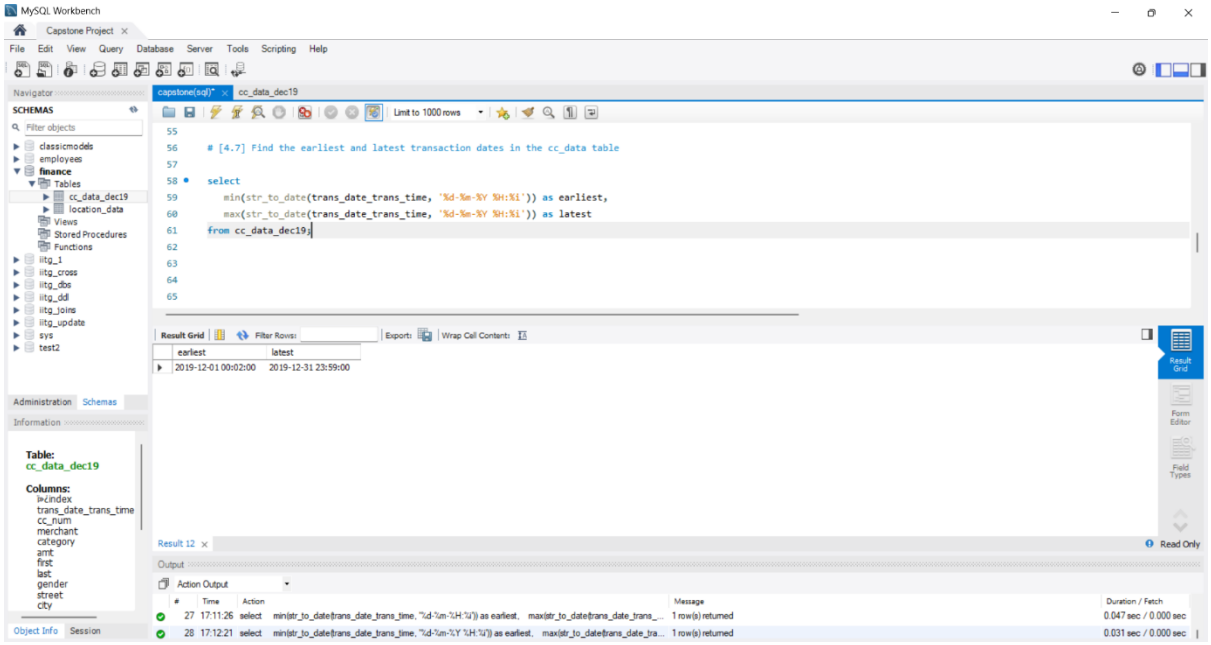
2.5. Join the **cc_data_dec19** and **location_data** tables to identify the latitude and longitude of each transaction



2.6. Identify the city with the highest population in the **location_data** table

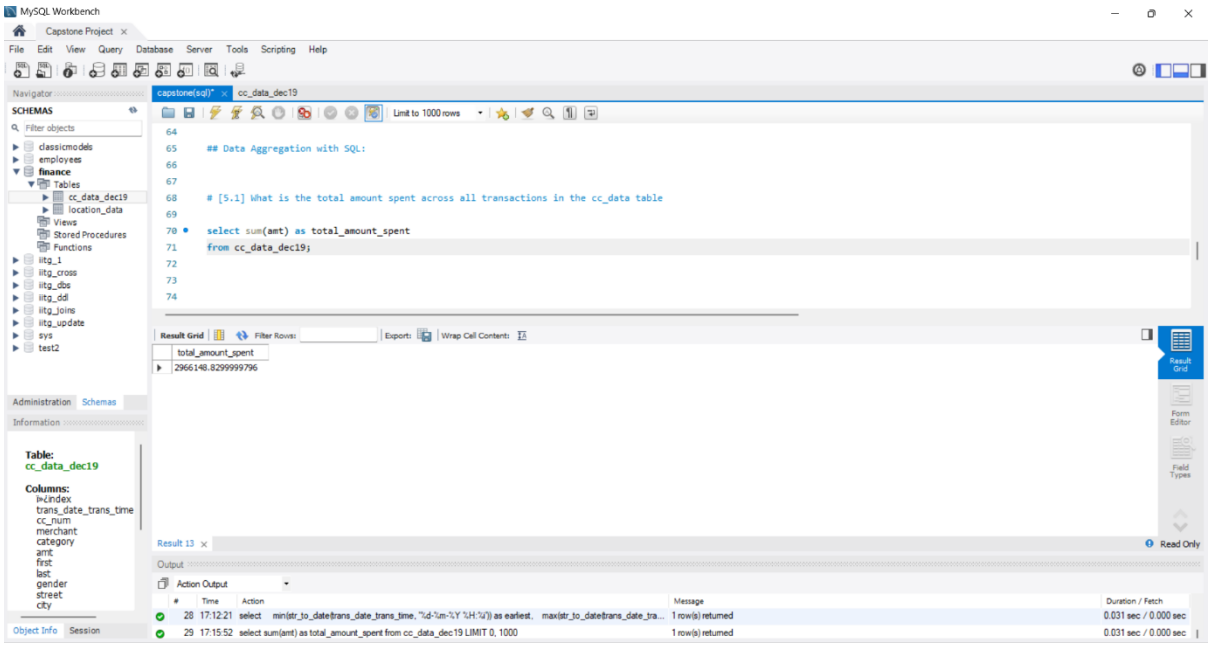


2.7. Find the earliest and the latest transaction dates in the **cc_data_dec19** table

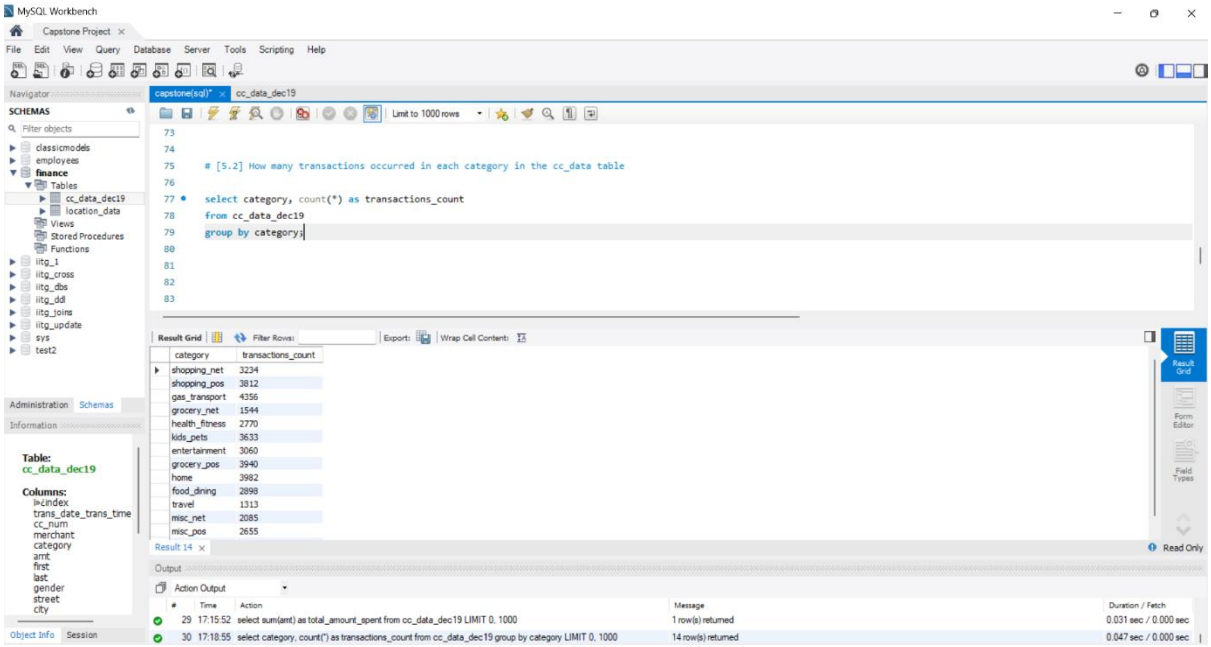


3. Data Aggregation with SQL:

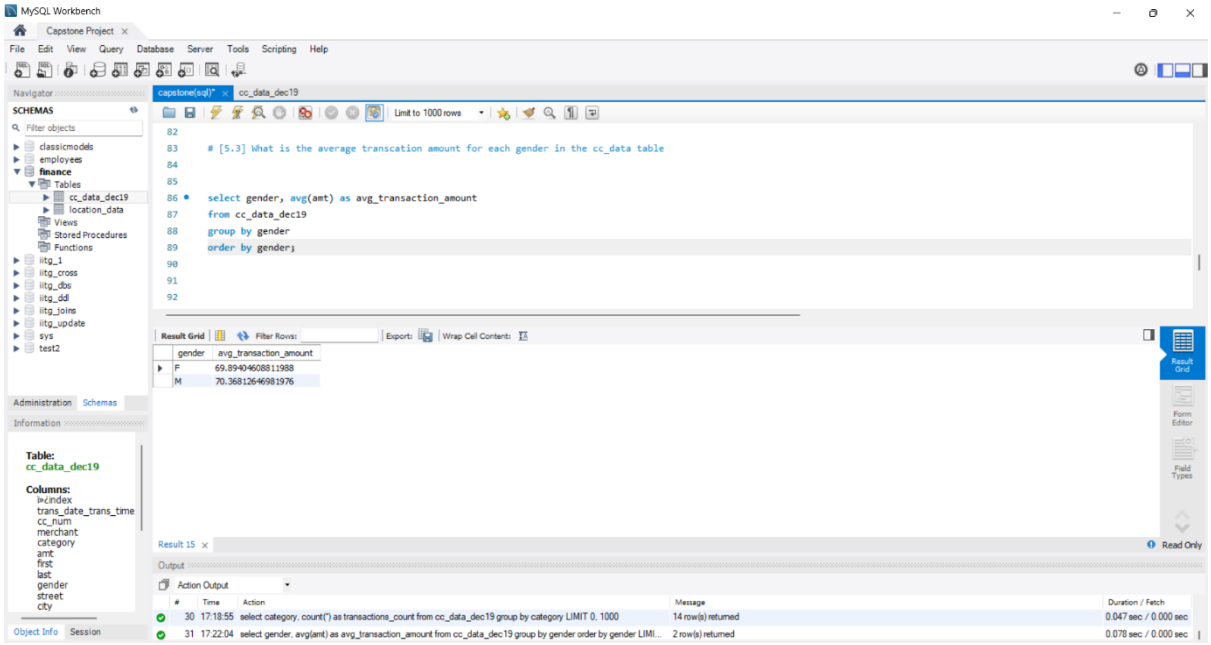
3.1. What is the total amount spent across all transactions in the **cc_data_dec19** table?



3.2. What is the total amount spent across all transactions in the **cc_data_dec19** table?



3.3. What is the average transaction amount for each gender in the **cc_data_dec19** table?



3.4. Which day of the week has the highest average transaction amount in the **cc_data_dec19** table?

MySQL Workbench

Capstone Project

File Edit View Query Database Server Tools Scripting Help

capstone(sql)* cc_data_dec19

Limit to 1000 rows

```
# [5.4] Which day of the week has the highest average transaction amount in the cc_data table

select
   dayname(str_to_date(trans_date_trans_time, '%d-%m-%Y %H:%i')) as day_of_week,
   avg(amt) as avg_transaction_amount
from cc_data_dec19
group by day_of_week
order by avg_transaction_amount desc
limit 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

day_of_week	avg_transaction_amount
Monday	74.2374315029878 6

Result 15

Output

Action Output

#	Time	Action	Message	Duration / Fetch
32	17:26:37	select dayname(str_to_date(trans_date_trans_time, '%d-%m-%Y %H:%i')) as day_of_week, avg(amt) as avg_tr...	Error Code: 1055. Expression #1 of SELECT list is not in GROUP BY clause and contains nonaggregated column...	0.016 sec
33	17:28:03	select dayname(str_to_date(trans_date_trans_time, '%d-%m-%Y %H:%i')) as day_of_week, avg(amt) as avg_tr...	1 row(s) returned	0.031 sec / 0.000 sec

Navigator

SCHEMAS

Filter objects

- classicmodels
- employees
- finance
  - Tables
    - cc_data_dec19
    - location_data
  - Views
  - Stored Procedures
  - Functions
- iitg_1
- iitg_cross
- iitg_dbs
- iitg_ddl
- iitg_joins
- iitg_update
- sys
- test2

Administration Schemas

Information

Table: cc_data_dec19

Columns:
- index
- trans_date_trans_time
- cc_num
- merchant
- category
- amt
- first
- last
- gender
- street
- city

Object Info Session